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The long read

No god in the machine: the pitfalls of AI worship

The rise of artificial intelligence has sparked a panic about computers gaining

power over humankind. But the real threat comes from falling for the hype

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Composite: Getty Images/Guardian design

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In Arthur C Clarke's short story *The Nine Billion Names of God*, a sect of monks in Tibet believes humanity has a divinely inspired purpose: inscribing all the various names of God. Once the list was complete, they thought, he would bring the universe to an end. Having worked at it by hand for centuries, the monks decide to employ some modern technology. Two sceptical engineers arrive in the Himalayas, powerful computers in tow. Instead of 15,000 years to write out all the permutations of God's name, the job gets done in three months. As the engineers ride ponies down the mountainside, Clarke's tale ends with one of literature's most economical final lines: "Overhead, without any fuss, the stars were going out."

It is an image of the computer as a shortcut to objectivity or ultimate meaning - which also happens to be, at least part of, what now animates the fascination with artificial intelligence. Though the technologies that underpin AI have existed for some time, it's only since late 2022, with the emergence of OpenAI's ChatGPT, that the technology that approached intelligence appeared to be much closer. In a [2023 report by Microsoft Canada](#), president Chris Barry proclaimed that "the era of AI is here, ushering in a transformative wave with potential to touch every facet of our lives", and that "it is not just a technological advancement; it is a societal shift". That is among the more level-headed reactions. Artists and writers are panicking that they will be made obsolete, governments are scrambling to catch up and regulate, and academics are debating furiously.

Businesses have been eager to rush aboard the hype train. Some of the world's largest companies, including Microsoft, Meta and Alphabet, are throwing their full weight behind AI. On top of the billions spent by big tech, [funding for AI startups hit nearly \\$50bn](#) in 2023. At an event at Stanford University in April, OpenAI CEO Sam Altman said he didn't really care if the company spent \$50bn a year on AI. Part of his vision is for a kind of super-assistant, one that would be a "super-competent

colleague that knows absolutely everything about my whole life, every email, every conversation I've ever had, but doesn't feel like an extension".

But there is also a profound belief that AI represents a threat. The philosopher **Nick Bostrom** is among the most prominent voices asserting that AI poses an existential risk. As he laid out in his 2014 book *Superintelligence*, if "we build machine brains that surpass human brains in general intelligence ... the fate of our species would depend on the actions of the machine super-intelligence." The classic cautionary tale here is that of an AI system whose only - seemingly inoffensive - goal is making paperclips. According to Bostrom, the system would realise quickly that humans are a barrier to this task, because they might switch off the machine. They might also use up the resources needed for the manufacturing of more paperclips. This is an example of what AI doomers call the "control problem": the fear that we will lose control of AI because any defences we've built into it will be undone by an intelligence that's millions of steps ahead of us.

Before we do, in fact, cede any more ground to our tech overlords, it's worth casting your mind back to the mid-1990s and the arrival of the world wide web. That, too, came with profound assertions of a new utopia, a connected world in which borders, difference and privation would end. Today, you would be hard pressed to argue that the internet has been some sort of unproblematic good. The fanciful did come true; we can carry the whole world's knowledge in our pockets. This just had the rather strange effect of driving people a bit mad, fostering discontent and polarisation, assisting a renewed surge of the far right and destabilising democracy and truth.

It's not that you should simply resist technology; it can, after all, also have liberating effects. Rather, when big tech comes bearing gifts, you should probably look closely at what's in the box.

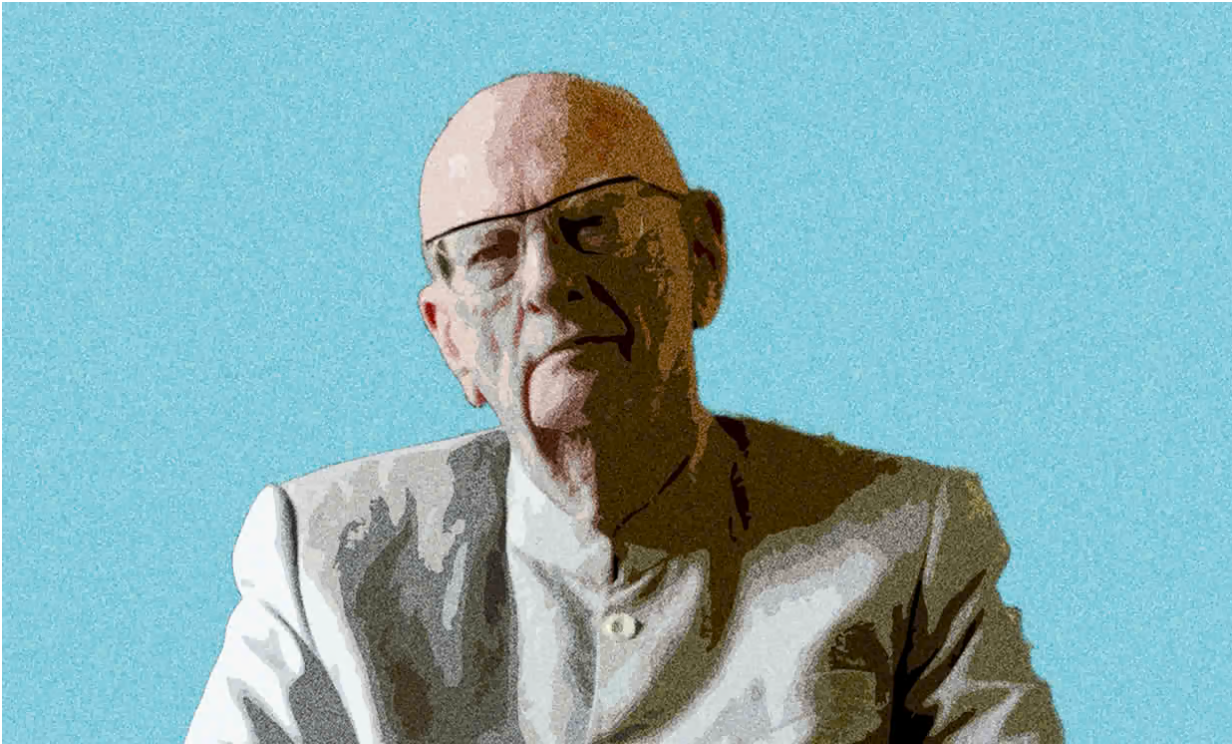
What we call AI at the moment is predominantly focused on large language models, or LLMs. The models are fed massive sets of data - **ChatGPT** essentially hoovered up the entire public internet - and trained to find patterns in them. Units of meaning, such as words, parts of words and characters, become tokens and are assigned numerical values. The models learn

how tokens relate to other tokens and, over time, learn something like context: where a word might appear, in what order, and so on.

That doesn't sound impressive on its own. But when I recently asked ChatGPT to write a story about a sentient cloud who was sad the sun was out, the results were strikingly human. Not only did the chatbot produce the various components of a children's fable, it also included a story arc in which, eventually, "Nimbus" the cloud found a corner of the sky and made peace with a sunny day. You might not call the story good, but it would probably entertain my five-year-old nephew.

Robin Zebrowski, professor and chair of cognitive science at Beloit College in Wisconsin, explains the humanity I sensed this way: "The only truly linguistic things we've ever encountered are things that have minds. And so when we encounter something that looks like it's doing language the way we do language, all of our priors get pulled in, and we think: 'Oh, this is clearly a minded thing.'"

This is why, for decades, the standard test for whether technology was approaching intelligence was the Turing test, named after its creator [Alan Turing](#), the British mathematician and second world war code-breaker. The test involves a human interrogator who poses questions to two unseen subjects - a computer and another human - via text-based messages to determine which is the machine. A number of different people play the roles of interrogator and respondent, and if a sufficient proportion of interviewers is fooled, the machine could be said to exhibit intelligence. ChatGPT can already fool at least some people in some situations.



The science fiction author Arthur C Clarke. Composite: Gemunu Amarasinghe/AP/Guardian design

Such tests reveal how closely tied to language our notions of intelligence are. We tend to think that beings that can “do language” are intelligent: we marvel at dogs that appear to understand more complex commands, or gorillas that can communicate in sign language, precisely because such acts are closer to our mechanism of rendering the world sensible.

But being able to do language without also thinking, feeling, willing or being is probably why writing done by AI chatbots is so lifeless and generic. Because LLMs are essentially looking at massive sets of patterns of data and parsing how they relate to one another, they can often spit out perfectly reasonable-sounding statements that are wrong or nonsensical or just weird. That reduction of language to just collection of data is also why, for example, when I asked ChatGPT to write a bio for me, it told me I was born in India, went to Carleton University and had a degree in journalism – about which it was wrong on all three counts (it was the UK, York University and English). To ChatGPT, it was the shape of the answer, expressed confidently, that was more important than the content, the right pattern mattering more than the right response.

All the same, the idea of LLMs as repositories of meaning that are then recombined does align with some assertions from 20th-century

philosophy about the way humans think, experience the world, and create art. The French philosopher Jacques Derrida, building on the work of linguist Ferdinand de Saussure, suggested that meaning was differential - the meaning of each word depends on that of other words. Think of a dictionary: the meaning of words can only ever be explained by other words, which in turn can only ever be explained by other words. What is always missing is some sort of “objective” meaning outside this neverending chain of signification that brings it to a halt. We are instead forever stuck in this loop of difference. Some, like Russian literary scholar Vladimir Propp, theorised that you could break down folklore narratives into constituent structural elements, as per his seminal work, *Morphology of the Folktale*. Of course, this doesn’t apply to all narratives, but you can see how you might combine units of a story - a starting action, a crisis, a resolution and so on - to then create a story about a sentient cloud.

Today, AI can take previously unconnected, even random things, such as the skyline of Toronto and the style of the impressionists, and join them to create what hasn’t existed before. But there is a sort of discomfoting or unnerving implication here. Isn’t that also, in a way, how we think? Raphaël Millière, an assistant professor at Macquarie University in Sydney, says that, for example, we know what a pet is (a creature we keep with us at home) and we also know what a fish is (an animal that swims in large water bodies); we combine those two in a way that keeps some characteristics and discards others to form a novel concept: a pet fish. Newer AI models boast this capacity to amalgamate into the ostensibly new - and it is precisely why they are called “generative.”

Even comparatively sophisticated arguments can be seen to work this way. The problem of theodicy has been a topic of debate among theologians for centuries. It asks: if an absolutely good God is omniscient, omnipotent and omnipresent, how can evil exist when God knows it will happen and can stop it? It radically oversimplifies the theological issue, but theodicy, too, is in some ways a kind of logical puzzle, a pattern of ideas that can be recombined in particular ways. I don’t mean to say that AI can solve our deepest epistemological or philosophical questions, but it does suggest that the line between thinking beings and pattern recognition machines is not quite as hard and bright as we may have hoped.

The sense of there being a thinking thing behind AI chatbots is also driven by the now common wisdom that we don't know exactly how AI systems work. What's called the black box problem is often framed in mystical terms - the robots are so far ahead or so alien that they are doing something we can't comprehend. That is true, but not quite in the way it sounds. New York University professor Leif Weatherby suggests that the models are processing so many permutations of data that it is impossible for a single person to wrap their head around it. The mysticism of AI isn't a hidden or inscrutable mind behind the curtain; it's to do with scale and brute power.

Yet, even in that distinction - that AI is able to do language only through computing power - there is still an interesting question of what it means to think. York University professor Kristin Andrews, who researches animal intelligence, suggests that there are lots of cognitive tasks - remembering how to get food, recognising objects or other beings - that animals do without necessarily being self-aware. In that sense, intelligence may well be attributed to AI because it can do what we would usually refer to as cognition. But, as Andrews notes, there's nothing to suggest that AI has an identity or a will or desires.

So much of what produces will and desire is located in the body, not just in the obvious sense of erotic desire but the more complex relation between an interior subjectivity, our unconscious, and how we move as a body through the world, processing information and reacting to it. Zebrowski suggests there is a case to be made that "the body matters for how we can think and why we think and what we think about". She adds, "It's not like you can just take a computer program and stick it in the head of a robot and have an embodied thing."

Computers might in fact approach what we call thinking, but they don't dream, or want, or desire, and this matters more than AI's boosters let on. When we use our intelligence to craft solutions to economic crises or to tackle racism, we do so out of a sense of morality, of obligation to those around us - our cultivated sense that we have a responsibility to make things better in specific, morally significant ways.

Perhaps the model of the computer in Clarke's story - something that is kind of a shortcut to transcendence or omniscience - is thus the wrong one. Instead, Deep Thought, the computer in Douglas Adams's *The Hitchhiker's Guide to the Galaxy*, might be closer. When asked "the

answer to the Ultimate Question of Life, the Universe and Everything”, it of course spits out that famously obtuse answer: “42”.

The absurdity of the answer points to an easily forgotten truth. Life and its meaning can’t be reduced to a simple statement, or to a list of names, just as human thought and feeling can’t be reduced to something articulated by what are ultimately ones and zeros. If you find yourself asking AI about the meaning of life, it isn’t the answer that’s wrong. It’s the question.

This March, I spent two days at the Microsoft headquarters, just outside Seattle. Microsoft is one of the tech incumbents that is most “all in” on AI. To prove just how much, they brought in journalists from around the world to participate in an “innovation campus tour”, which included a dizzying run of talks and demos, meals in the seemingly neverending supply of on-site restaurants, and a couple of nights in the kind of hotel writers usually cannot afford.

We were walked through a research centre, wearing earplugs to block out the drone of a mini field of fans. We attended numerous panels: how teams were integrating AI, how those involved with “responsible AI” are tasked with reining in the technology before it goes awry. There was lots of chatter about how this work was the future of everything. In one session, the affable Seth Juarez, principal programme manager of AI platforms, spoke of AI as being like moving from a shovel to a tractor: that it will, in his words, “level up humanity”.

Some of the things we saw were genuinely inspiring, such as the presentation by Saqib Shaikh, who is blind and has spent years working on SeeingAI. It’s an app that is getting better and better at labelling objects in a field of view in real time. Point it at a desk with a can and it will say, “A red soda can, on a green desk.” There were other hopeful ideas for how AI could be used: to preserve dying languages, improve the accuracy of scans for tumours, or better predict where to deploy disaster response resources – usually by processing large amounts of data and then recognising and analysing patterns within it.

At the end of a panel at Microsoft on AI research, we were each offered a copy of a book titled AI for Good, detailing more altruistic uses. One of

the projects mentioned in the book involved using machine learning to predict where best to put solar panels in India. That's encouraging stuff, the sort of thing that lets one momentarily feel a hint of relief or hope that, maybe, some things are going to get better. But many of the problems preventing, say, the deployment of solar power in India do not relate to a lack of knowledge. More often, the problems are the lack of resources, the absence of political will, the power of entrenched interests and, more plainly, money.

This is what the utopian vision of the future so often misses: if and when change happens, the questions at play will be about if and how certain technology gets distributed, deployed, taken up. It will be about how governments decide to allocate resources, how the interests of various parties affected will be balanced, how an idea is sold and promulgated, and more. It will, in short, be about political will, resources, and the contest between competing ideologies and interests. The problems facing the world - not just climate breakdown but the housing crisis, the toxic drug crisis, or growing anti-immigrant sentiment - aren't problems caused by a lack of intelligence or computing power. In some cases, the solutions to these problems are superficially simple. Homelessness, for example, is reduced when there are more and cheaper homes. But the fixes are difficult to implement because of social and political forces, not a lack of insight, thinking, or novelty. In other words, what will hold progress on these issues back will ultimately be what holds everything back: us.

The idea of an exponentially greater intelligence, so favoured by big tech, is a strange sort of fantasy that abstracts out intelligence into a kind of superpower that can only ever increase. In this view, problem-solving is like a capacity on a dial that can simply be turned up and up. To assume this is what's called "**tech solutionism**", a term coined a decade ago by the writer Evgeny Morozov. He was among the first to point to how Silicon Valley tended to see tech as the answer to everything.

Computer pioneer Alan Turing. Composite: Universal Images Group/Getty Images/Guardian design

Some Silicon Valley businessmen have taken tech solutionism to an extreme. It is these AI **accelerationists** whose ideas are the most terrifying. Marc Andreessen was intimately involved in the creation of the first web browsers and is now a billionaire venture capitalist who has taken up a mission to fight against the “woke mind virus” and generally embrace capitalism and libertarianism. In a screed published last year, titled The Techno-Optimist Manifesto, Andreessen outlined his belief that “there is no material problem - whether created by nature or by technology - that cannot be solved with more technology”. When the historian Rick Perlstein attended a dinner at Andreessen’s \$34m home in California, he found a group adamantly opposed to regulation or any kind of constraint on tech (in a tweet at the end of 2023, Andreessen called regulation of AI “the new foundation of totalitarianism”). When Perlstein related the whole experience to a colleague, he “noted a similarity to a student of his who insisted that all the age-old problems historians worried over would soon obviously be solved by better computers, and thus considered the entire humanistic enterprise faintly ridiculous”.

Andreessen’s manifesto also included a section in which he listed off a series of enemies. It included all the usual rightwing bugbears: regulation, know-it-all academics, the constraint on “innovation”,

progressives themselves. To the venture capitalist, these are all self-evident evils. Andreessen has been on the board of Facebook/Meta - a company that has allowed mis- and disinformation to wreak havoc on democratic institutions - since 2008. However, he insists, apparently without a trace of irony, that experts are “playing God with everyone else’s lives, with total insulation from the consequences”.

Having spoken to experts, it seems to me as though the promise of AI lies in dealing with sets of data that exist at a scale humans simply cannot operate at. Pattern recognition machines put to use in biology or physics will probably yield fascinating, useful results. AI’s other uses seem more mundane, at least for now: taking financial statements and reconciling figures, transcribing and summarising meetings, triaging emails more efficiently, making it easier to access simple things like travel itineraries.

That isn’t to say AI is some benevolent good, however. An AI model can be trained on billions of data points, but it can’t tell you if any of those things is good, or if it has value to us, and there’s no reason to believe it will. We arrive at moral evaluations not through logical puzzles but through consideration of what is irreducible in us: subjectivity, dignity, interiority, desire - all the things AI doesn’t have.

To say that AI on its own will be able to produce art misunderstands why we turn to the art in the first place. We crave things made by humans because we care about what humans say and feel about their experience of being a person and a body in the world.

There’s also a question of quantity. In dropping the barriers to content creation, AI will also flood the world with dreck. Already, Google is becoming harder to use because the web is being flooded with AI-crafted content designed to get clicks.

Yet it’s also the imbrication of AI into existing systems that is cause for concern. As Damien P Williams, a professor at the University of North Carolina at Charlotte, pointed out to me, training models absorb masses of data based on what is and what has been. It’s thus hard for them to avoid existing biases, of the past and the present. Williams points to how, if asked to reproduce, say, a doctor yelling at a nurse, AI will make

the doctor a man and the nurse a woman. Last year, when Google hastily released Gemini, its competitor to other AI chatbots, it produced images of “diverse” Nazis and the US’s founding fathers. These odd mistakes were a ham-fisted attempt to try to pre-empt the problem of bias in the training data. AI relies on what has been, and trying to account for the myriad ways we encounter and respond to the prejudice of the past appears to simply be beyond its ken.

OpenAI CEO Sam Altman. Composite: Carlos Barría/Reuters/Guardian design

The structural problem with bias has existed for some time. Algorithms were already used for things like credit scores, and already AI usage in things like hiring is replicating biases. In both cases, pre-existing racial bias emerged in digital systems. That’s not to say that AI won’t also kill us. More recently, it was revealed that Israel was using a version of AI called Lavender to help it attack targets in Gaza. The system is meant to identify members of Hamas and Palestinian Islamic jihad and then provide their locations as potential targets for airstrikes - including their homes. According to the Israeli Palestinian [+972 Magazine](#), many of these attacks killed civilians.

As such, the threat of AI isn’t really that of a machine or system which offhandedly kills humanity. It’s the assumption that AI is in fact intelligent that causes us to outsource crucial social and political functions to computer software - it’s not just the tech itself which becomes integrated into day-to-day life but also the particular logic and

ethos of tech and its libertarian-capitalist ideology.

The question is then: to what ends AI is deployed, in what context, and with what boundaries. “Can AI be used to make cars drive themselves?” is an interesting question. But whether we should allow self-driving cars on the road, under what conditions, embedded in what systems – or indeed, whether we should deprioritise the car altogether – are the more important questions, and they are ones that an AI system cannot answer for us.

Everyone from a tech bro placing his hopes for human advancement on a superhuman intelligence to an army relying on AI software to list targets evinces the same desire for an objective authority figure to which one can turn. When we look to artificial intelligence to make sense of the world – when we ask it questions about reality or history or expect it to represent the world as it is – are we not already bound up in the logic of AI? We are awash with digital detritus, with the cacophony of the present, and in response, we seek out a superhuman assistant to draw out what is true from the morass of the false and the misleading – often to only be misled ourselves when AI gets it wrong.

We are living in a time where truth is unstable, shifting, constantly in contestation. Think of the embrace of conspiracy theories, the rise of the anti-vax movement, or the mainstreaming of racist pseudoscience. Every age has its great loss – for modernism, it was the coherence of the self; for postmodernism, the stability of master narratives – and now, in the 21st century, there is an increasing pressure on the notion of a shared vision of reality.

When the systems that give shape to things start to fade or come into doubt, as has happened to religion, liberalism, democracy and more, one is left looking for a new God. There is something particularly poignant about the desire to ask ChatGPT to tell us something about a world in which it can occasionally feel like nothing is true. To humans awash with a sea of subjectivity, AI represents the transcendent thing: the impossibly logical mind that can tell us the truth.

Lingering at the edges of Clarke’s short story about the Tibetan monks

was a similar sense of technology as the thing that lets us exceed our mere mortal constraints. But the result is the end of everything. In turning to technology to make a deeply spiritual, manual, painstaking task more efficient, Clarke's characters end up erasing the very act of faith that sustained their journey toward transcendence. But here in the real world, perhaps meeting God isn't the aim. It's the torture and the ecstasy of the attempt to do so. Artificial intelligence may keep growing in scope, power and capability, but the assumptions underlying our faith in it - that, so to speak, it might bring us closer to God - may only lead us further away.

In 10 or 20 years from now, AI will undoubtedly be more advanced than it is now. Even so, if I'm lucky enough to be around, I'll step out of my home with my AI assistant whispering in my ear. There will still be cracks in the sidewalk. The city in which I live will still be under construction. Traffic will probably still be a mess, even if the cars drive themselves. Maybe I'll look around, or look up at the sky, and my AI assistant will tell me something about what I see. But things will still keep moving on only slightly differently than they do now. And the stars? Against what might now seem like so much change, the sky will still be full of them.

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